

SKILLS

- **Quantitative & Research:** Statistical modelling, time-series analysis, hypothesis testing, experiment design, backtesting, Monte Carlo simulation, performance monitoring, Python (Pandas, NumPy, SciPy), R, MATLAB
- **ML & Optimization:** PyTorch, TensorFlow, Scikit-learn, reinforcement learning (DQN/DDQN), regression, classification, portfolio optimization, anomaly detection
- **Data & Pipelines:** SQL, Kafka, Airflow, Snowflake, Elasticsearch, Redis, OpenTelemetry, ELK Stack, ETL, schema design, real-time streaming
- **Languages & Tools:** Python, Go, TypeScript, JavaScript, SQL | Git, Jupyter, Tableau
- **Cloud & Platforms:** AWS (Kinesis, Lambda, SageMaker, S3), Docker, Kubernetes, Terraform, CI/CD

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect Nov 2022 – Present
 - **Logging as a Service (LAS):** Architected institution-wide log repatriation platform (Vector, Fluent Bit, Kafka, Logstash) ingesting 150 TB/day across 10,000+ systems; defined OpenTelemetry log and metric schemas; enabled vendor-agnostic observability switching (Dynatrace, Datadog, Splunk) as a configuration change; eliminated point-to-point third-party integrations institution-wide.
 - **M365 Log Repatriation:** Designed custom collector repatriating Microsoft 365 audit logs (Exchange, SharePoint) at 60 TB/day with geolocation tagging; removed Microsoft data-custody dependencies and standardised high-volume pipeline monitoring for operational analytics.
 - **OpenTelemetry Standardisation:** Drove vendor-agnostic OTel adoption as institutional standard; authored collection pipelines, schema contracts, and SDK onboarding guides adopted by 20+ engineering teams; standardised formats for AIOps and operational analytics use cases.
 - **ASB Builder (Patent-Pending, Rules-Based Engine):** Led 0-to-1 delivery of enterprise architecture blueprint platform (Node.js/TypeScript, React) with pattern-based recommendation engine, auto-certification workflow, and real-time data rendering; maps 1,000+ architecture patterns to component selections—rules- and data-driven, not ML-based.
 - **Technical Reference Model & Team Leadership:** Digitalised RBC technical reference model (1,000+ technologies, 1,500+ capabilities); led 5 engineers with sprint delivery, code reviews, and ADRs; partnered with product, risk, and architecture stakeholders in Enterprise Architecture (Athena) certification workflows.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - **Insurance Validation Algorithm:** Designed proprietary validation and trust-score logic on Hyperledger Fabric for real-time premium verification and fraud-resistant claims processing; joined as first engineer and scaled platform to 20+ engineers before acquisition.
 - **ML Document Classification:** Integrated transformer-based models for medical record parsing on AWS EKS with HIPAA-compliant audit logging; reduced manual review overhead by 60% and enabled real-time credential verification at scale.
- **Dalhousie University** Halifax, Canada
Research & Teaching Assistant Jan 2021 – Aug 2022
 - **Compiler/Runtime Research & Instruction:** Research assistant on call-stack decision algorithms and Python tracing-table tooling for compiler/runtime research; TA for graduate Software Development & Algorithms and Advanced Cloud Computing plus undergraduate software development labs.
- **CryptoVantage** Netherlands (Remote)
Blockchain Developer Jul 2020 – Dec 2020
 - **Illicit Transaction Detection Engine:** Built real-time on-chain analytics in Go/JavaScript with Ethereum event streaming, ML-based wallet clustering, and risk scoring for 5AMLD (EU Anti-Money Laundering Directive) compliance workflows.

PROJECTS

- **Automatic Trading Algorithm (Personal):** Sub-second latency personal project integrating live brokerage APIs and real-time market data streams; momentum and mean-reversion quantitative signals with automated risk management and position sizing.
- **Flight Price Optimization Agent (Personal):** Reinforcement learning agent (DQN/DDQN, Python) for dynamic pricing optimisation—modelled demand curves on historical flight data with 18% improvement over baseline in simulation.
- **DCS-BBN (Decentralised Cloud Storage):** Peer-to-peer archival storage using Solidity smart contracts, Hyperledger Fabric ledger, and zk-SNARK proofs. **Published – Springer ICMR International Conference on Mobile Computing.**

EDUCATION

- **Dalhousie University** Halifax, Canada
MASc Applied Computer Science, GPA: 3.9/4.0 | TA: Algorithms, Advanced Cloud Computing Jan 2021 – Aug 2022
- **Gujarat Technological University** Ahmedabad, India
B.E. Computer Engineering, GPA: 3.92/4.0 Aug 2016 – Aug 2020